

CPEN 416

Lecture 07: superdense coding and quantum teleportation

Thursday 1 October 2026

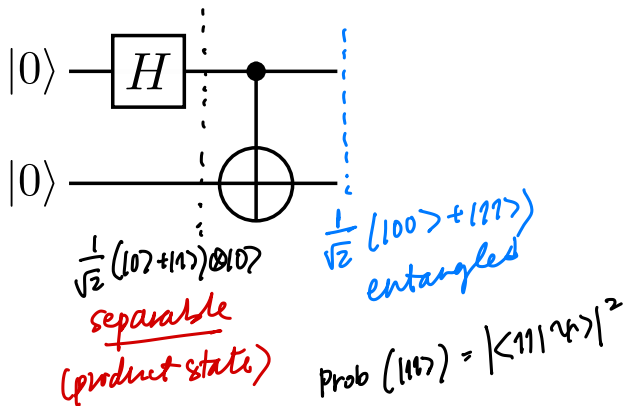
Announcements

- Tutorial assignment 1 due tomorrow 23:59
- Bring me your midterm reference sheets for approval by 5pm tomorrow
- Monday's tutorial: midterm prep

★ suggest practice midterm questions on Piazza

Last time

We introduced the idea of entanglement and entangling gates.



Learning outcomes

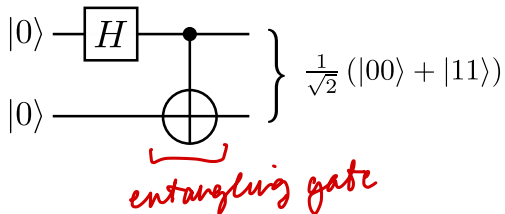
- measure multi-qubit systems in the Bell basis
- leverage entanglement to implement superdense coding
- prove that arbitrary quantum states cannot be “cloned”
- teleport a state from one qubit to another

Bell states

Remember how we created

$$|\psi_{00}\rangle = \frac{1}{\sqrt{2}} (|00\rangle + |11\rangle),$$

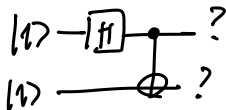
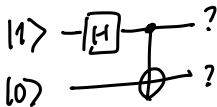
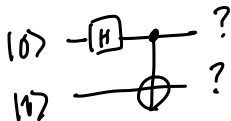
from the $|00\rangle$ state:



Bell states

$$|00\rangle \rightarrow \frac{1}{\sqrt{2}} (|00\rangle + |11\rangle)$$

Exercise: Apply the same circuit to the other 3 computational basis states? What is special about these four states?



$$\frac{1}{\sqrt{2}} (|01\rangle + |10\rangle)$$

$$\frac{1}{\sqrt{2}} (|01\rangle - |10\rangle) \xrightarrow{\text{CNOT}} \frac{1}{\sqrt{2}} (|00\rangle - |11\rangle)$$

$$\frac{1}{\sqrt{2}} (|01\rangle - |10\rangle)$$

all entangled!!

all mutually orthogonal $\langle \Psi_{10} | \Psi_{01} \rangle = ? = 0$

$$\begin{aligned} |\Psi_{00}\rangle &= \frac{1}{\sqrt{2}} (|00\rangle + |11\rangle) & |\Psi_{01}\rangle &= \frac{1}{\sqrt{2}} (|01\rangle + |10\rangle) & \text{BELL BASIS} \\ |\Psi_{10}\rangle &= \frac{1}{\sqrt{2}} (|00\rangle - |11\rangle) & |\Psi_{11}\rangle &= \frac{1}{\sqrt{2}} (|01\rangle - |10\rangle) \end{aligned}$$

Review: basis rotations

Measure in basis $|\psi_1\rangle, |\psi_2\rangle$

find mapping $\begin{matrix} |0\rangle \rightarrow |\psi_1\rangle \\ |1\rangle \rightarrow |\psi_2\rangle \end{matrix} \} U$

$\text{---} [C] \text{---} [X] \{|\psi_1\rangle, |\psi_2\rangle\} \Rightarrow \begin{matrix} U^\dagger |\psi_1\rangle = |0\rangle \\ U^\dagger |\psi_2\rangle = |1\rangle \end{matrix}$

But basis measurement?? $\text{---} [C] \text{---} [U^\dagger] \text{---} [X] \begin{matrix} |0\rangle \Rightarrow |\psi_1\rangle \\ |1\rangle \Rightarrow |\psi_2\rangle \end{matrix}$
basis rotation

$\text{---} [C] \text{---} [X] \{|\psi_0\rangle, |\psi_1\rangle, \dots\} \Rightarrow \text{---} [C] \text{---} [H] \text{---} [X]$

Bell basis measurement

$$\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle) \left\{ \begin{array}{c} \text{---} \bullet \text{---} [H] \text{---} |0\rangle \\ | \\ \text{---} \oplus \text{---} |0\rangle \end{array} \right.$$

$$\frac{1}{\sqrt{2}}(|01\rangle + |10\rangle) \left\{ \begin{array}{c} \text{---} \bullet \text{---} [H] \text{---} |0\rangle \\ | \\ \text{---} \oplus \text{---} |1\rangle \end{array} \right.$$

$$\frac{1}{\sqrt{2}}(|00\rangle - |11\rangle) \left\{ \begin{array}{c} \text{---} \bullet \text{---} [H] \text{---} |1\rangle \\ | \\ \text{---} \oplus \text{---} |0\rangle \end{array} \right.$$

$$\frac{1}{\sqrt{2}}(|01\rangle - |10\rangle) \left\{ \begin{array}{c} \text{---} \bullet \text{---} [H] \text{---} |1\rangle \\ | \\ \text{---} \oplus \text{---} |1\rangle \end{array} \right.$$

Two famous quantum algorithms, **superdense coding** and **teleportation** work by performing a measurement in the Bell basis.

Superdense coding

Suppose Alice wants to send Bob two classical bits of information, say '1' and '0'.

Q1: How many classical bits must she send to Bob to do this?

2

Q2: How many *qubits* must she send to Bob to do this?

1

(if the qubit is part of a shared entangled pair?)

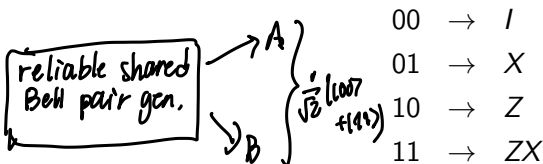
Superdense coding

Alice and Bob start the protocol with this shared entangled state:

$$|\Phi\rangle_{AB} = \frac{1}{\sqrt{2}} (|00\rangle + |11\rangle)$$

Alice's qubit?

Next, depending on her bits, Alice performs one of the following operations on her qubit:



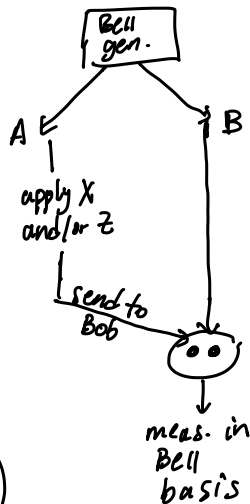
Superdense coding

What happened to the entangled state?

$$|\Phi\rangle_{AB} = \frac{1}{\sqrt{2}} (|00\rangle + |11\rangle)$$

It will transform to:

$$\begin{aligned} 00 &\rightarrow I & \frac{1}{\sqrt{2}} (|00\rangle + |11\rangle) \\ 01 &\rightarrow X & \frac{1}{\sqrt{2}} (|10\rangle + |01\rangle) \\ 10 &\rightarrow Z & \frac{1}{\sqrt{2}} (|00\rangle - |11\rangle) \\ 11 &\rightarrow ZX & \frac{1}{\sqrt{2}} (-|10\rangle + |01\rangle) \end{aligned}$$



Superdense coding

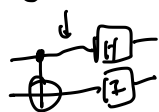
Now, Bob can either

- perform a measurement directly in the Bell basis
- perform a basis transformation from the Bell basis back to the computational basis

to determine his state with certainty, and thus, the bits Alice sent.



$$(H \otimes I)\text{CNOT} \cdot \frac{1}{\sqrt{2}} (|00\rangle + |11\rangle) = |00\rangle$$



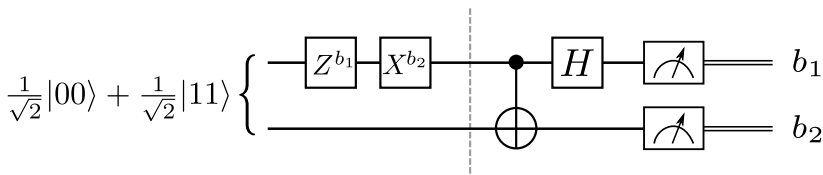
$$(H \otimes I)\text{CNOT} \cdot \frac{1}{\sqrt{2}} (|10\rangle + |01\rangle) = |01\rangle$$

$$(H \otimes I)\text{CNOT} \cdot \frac{1}{\sqrt{2}} (|00\rangle - |11\rangle) = |10\rangle$$

$$(H \otimes I)\text{CNOT} \cdot \frac{1}{\sqrt{2}} (|01\rangle - |10\rangle) = |11\rangle$$

Hands-on: superdense coding

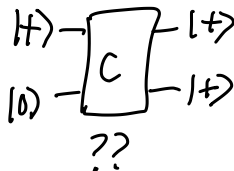
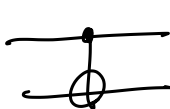
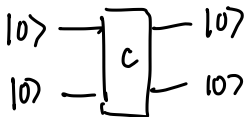
Let's go implement it!



Copying quantum states

Suppose you found a really cool quantum state, and you want to send a copy to a friend. Under what conditions is this possible?

Exercise: develop a circuit that “copies” $|0\rangle$ and $|1\rangle$.



Copying quantum states

Will your circuit work in general? Is there a state for which it fails?

$$\begin{array}{c} |+\rangle \\ |0\rangle \end{array} \begin{array}{c} \bullet \\ | \\ \oplus \end{array} = \begin{array}{c} |0\rangle \\ |0\rangle \end{array} \begin{array}{c} \boxed{H} \bullet \\ | \\ \oplus \end{array} \} \frac{1}{\sqrt{2}} (|100\rangle + |111\rangle)$$

$$\begin{aligned} \frac{1}{\sqrt{2}} (|100\rangle + |111\rangle) &\neq |+\rangle \otimes |+\rangle \\ &\neq \frac{1}{2} (|100\rangle + |101\rangle + |110\rangle + |111\rangle) \end{aligned}$$

(Not) copying quantum states

The no-cloning theorem

It is impossible to create a copying circuit that works for arbitrary quantum states.

In other words, there is no circuit that sends

$$U(|\psi\rangle \otimes |s\rangle) = |\psi\rangle \otimes |\psi\rangle$$

for any arbitrary $|\psi\rangle$. (or $|s\rangle$)

Proof of the no-cloning theorem

Suppose we want to clone a state $|\psi\rangle$. We want a unitary operation that sends

$$U(|\psi\rangle \otimes |s\rangle) = |\psi\rangle \otimes |\psi\rangle$$

where $|s\rangle$ is some arbitrary state.

Suppose we find one. If our cloning machine is to be universal, we must also be able to clone some other state, $|\varphi\rangle$.

$$U(|\varphi\rangle \otimes |s\rangle) = |\varphi\rangle \otimes |\varphi\rangle$$

Proof of the no-cloning theorem

We purportedly have:

$$\begin{aligned}
 U(|\psi\rangle \otimes |s\rangle) &= |\psi\rangle \otimes |\psi\rangle & \langle 10 | 11 \rangle &= \langle 11 | 01 \rangle \\
 U(|\varphi\rangle \otimes |s\rangle) &= |\varphi\rangle \otimes |\varphi\rangle & \langle 11 | 01 \rangle &= \langle 11 | 11 \rangle
 \end{aligned}$$

Take the inner product of the LHS of both equations:

$$\begin{aligned}
 \langle \psi | \otimes \langle s | \underbrace{U^\dagger \cdot U}_I (|\varphi\rangle \otimes |s\rangle) &= \langle \psi | \otimes \langle s | (|\varphi\rangle \otimes |s\rangle) \\
 &= \langle \psi | \varphi \rangle \cdot \langle s | s \rangle \\
 &= \langle \psi | \varphi \rangle
 \end{aligned}$$

Now take the inner product of the RHS of both equations:

$$\langle \varphi | \otimes \langle \varphi | \cdot |\psi\rangle \otimes |\psi\rangle = \langle \varphi | \psi \rangle \cdot \langle \varphi | \psi \rangle = \langle \varphi | \psi \rangle^2$$

Proof of the no-cloning theorem

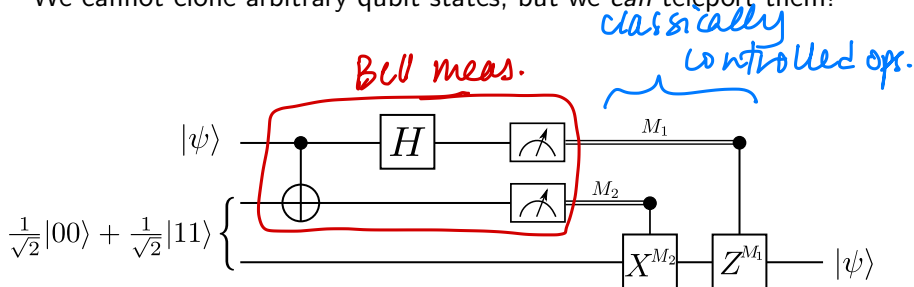
For what states does

$$(\langle\psi|\varphi\rangle)^2 = \langle\psi|\varphi\rangle$$

either $|\varphi\rangle = |\psi\rangle$
or orthogonal.

Teleportation

We cannot clone arbitrary qubit states, but we *can* teleport them!

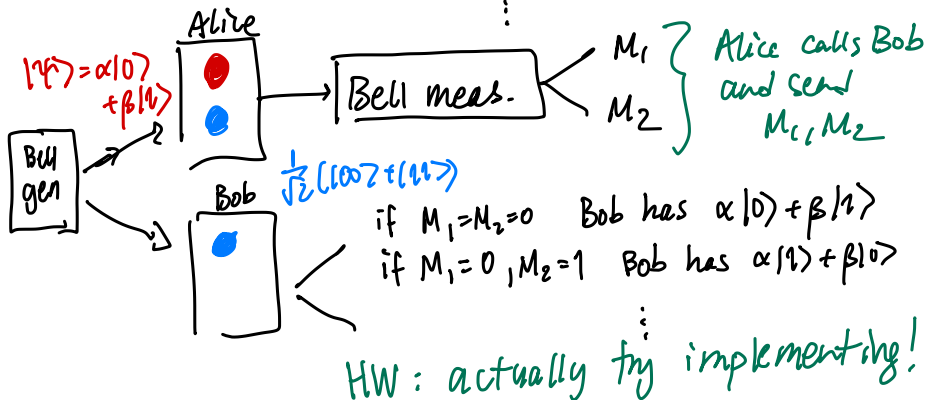
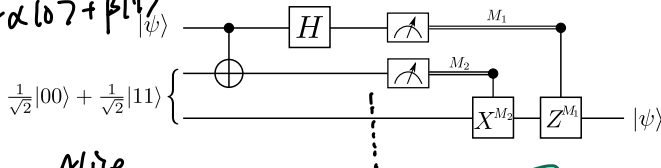


Exercise: determine the state of the system immediately before the measurements.

work through the full circuit once.

Teleportation

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$



Quantum teleportation: the details

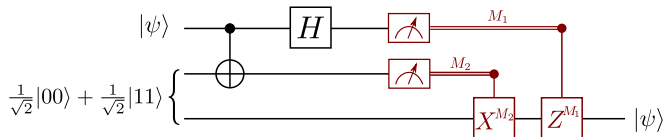
Before measurements, the combined state of the system is

What do you notice about this state?

Quantum teleportation: the details

Alice measures in the computational basis and sends her results to Bob, who can adjust his state as needed.

Let's implement it!



Next time

Content:

- Midterm on 6 October
- After midterm: Oracle-based algorithms (Deutsch and Grover)

Action items:

- 1 Submit tutorial assignment 1
- 2 Study for midterm (practice problems and midterms on PL)
- 3 Submit midterm reference sheet for review by Tuesday 2pm
- 4 Attend Monday's tutorial

Recommended reading (lecture 01-07, and midterm)

- Codebook nodes IQC, SQ, MQ
- Nielsen & Chuang 1.2, 1.3.1-1.3.7, 2.2.3-2.2.5, 2.3, 4.3

Read ahead: Codebook BA, GA; Nielsen & Chuang 1.4, 6.1